

Program : Diploma in Wood and Paper Technology	
Course Code : 3121	Course Title: Pulp Technology
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:2, T:1, P:0)	Periods per semester: 45

Course Objectives:

- To provide knowledge to the students regarding the basic concepts of pulping with details of physical and chemical characteristics of fibrous raw materials..
- To familiarize the students with various types of pulping methodologies used for pulp manufacture.

Course Prerequisites:

Topic	Course code	Course name	Semester
Structure of wood		Botany and anatomical structures	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Comprehend the preparation of raw materials for Pulping	11	Understanding
CO2	Summarise the types of pulping processes	10	Understanding
CO3	Describe the Washing , Screening and Cleaning of Pulp	11	Understanding
CO4	Comprehend the Bleaching of Pulp	11	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3				3		
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Comprehend the preparation of raw materials for Pulping		
M1.01	Describe the preparation of raw materials	3	Understanding
M1.02	Explain the fundamentals in characterization and classification of solids	4	Understanding
M1.03	Apply the fundamentals of conveying and storage	4	Understanding
Contents: Raw materials - sources - classification - selection - transportation - debarkers - chipping - drum chipper - disc chipper - calculation of chip size - screening of chips - screens - shaking screens - vibrating screens - grizzlies - bucket elevators - sieve analysis - re chipper - conveying of chips - belt conveyor - chain conveyors, roller conveyor, pneumatic conveyors scrap conveyor -screw conveyor - advantageous & disadvantageous - Chip storage - chip silo			
CO2	Summarise the types of pulping processes		
M2.01	Explain the classification of pulping process	3	Understanding
M2.02	Acquire knowledge of principles of Mechanical and semi-chemical pulping	4	Understanding
M2.03	Compare different pulping processes	3	Understanding
	Series Test - I	1	
Contents: Classification of pulping process - Chemical pulping - Soda pulping - sulphite process - sulphate pulping - bisulphite process - Principles of chemical pulping - soda process - sulphate process - sulphate pulping - bisulphate process - Bath ratio - Active alkali - effective alkali - sulphidity - consistency - TTAkappa number - permanganate number - cooking process - Digester - Classification - batch process - Direct & indirect heating - Continuous cooking - Kamyr - Pandya digester - rotary Digester			

Mechanical pulping and semi-chemical pulping-Principles of mechanical pulping - comparison of pulping processes. Types of mechanical pulping- Ground wood pulping - Grinders - types of grinders; Refiners - variables in refining. Thermomechanical pulping, Chemi thermomechanical pulping. Principles of semichemical pulping. Different semi-chemical pulping processes. Chemi-mech pulping. Super batch cooking. RDH cooking - Oxygen delignification.

CO3	Describe the Washing, Screening and Cleaning of Pulp		
M3.01	List the classification of secondary fibers	2	Understanding
M3.02	Explain Deinking and Depithing methods	3	Understanding
M3.03	Comprehend washing of pulp	3	Understanding
M3.04	Describe screening and cleaning of pulp	3	Understanding

Contents:

Definition - classification of secondary fibers - Production of secondary fibers. Importance of recycling. Deinked pulp - advantages - economics of deinking. Classification of ink - Deinking process. Variables in deinking operation. Flotation cell. Dipergent of deinked stock - properties and uses of deinked paper. Effluent treatment and waste disposal in deinking plant. Depithing - classification of depithing methods. Pulping of bagasse - properties and uses of bagasse pulp

Washing of pulp - objectives and principles. Rotary drum vacuum washer, pressure washer. High consistency washing - latest development in washing. Presses used in washing Screw press - disc press - belt press. Screening and cleaning of pulp. Need of screening and cleaning. Types of screens - rotary screen - pressure screen - centrifugal screen - high consistency screen. Centry cleaners - vertical and horizontal cleaners - inclined cleaner - high consistency cleaner

CO4	Comprehend the Bleaching of Pulp and importance of agricultural residue		
M 4.01	Explain the theory of bleaching	3	Understanding
M 4.02	Discuss the types of bleaching	2	Understanding
M 4.03	Summarise the Preparation of bleaching chemicals	3	Understanding
M 4.04	Outline the importance of agricultural residue	3	Understanding
	Series Test - II	1	

Contents:

Bleaching - need of bleaching - theory of bleaching - Various stages in bleaching - Types of chemicals - parameters - temperature, time, P^H , consistency.

Equipments in bleach plant - Types of mixers, Types of bleaching - chlorine, chlorine dioxide, hypo chlorate bleaching, Multi stage bleaching, oxygen bleaching, ozone bleaching - Total chlorine free bleaching, Elemental chlorine free bleaching. Preparation of bleaching chemicals. Productive bleaching.

Cooking of agricultural residue - importance of agricultural residue. Preparation and pulping of rags, rice, straw, cereal straw, esparto, hemp, jute, flax, bamboo, cotton linter.

Text / Reference:

T/R	Book Title/Author
T1	Ronald. G. Macdonald, Pulp & paper science Vol. I, II, - McGraw Hill Bookco, Newyork
R1	James.P.Casey, Pulp & paper chemistry and chemical technology Vol. I & II
R2	Britt ,Hand book of pulp & paper technology - CBS Publishers
R3	Swen.A.Rydholm, Pulping process
R4	Pulping process - Interscience publishers - John Viley& sons, Newyork
R5	Rance, Handbook of paper science Vol. I, II - H.E
R6	Recyling paper from fibre to finished product - TAPPI press
R7	Libby C.E., Pulp & paper science Vol. I, II, - McGraw Hill, Newyork
R8	TAPPI technical information sheets, I, II, III - TAPPI press

Online Resources:

Sl.No	Website Link
1	https://www.pulpandpaper-technology.com/